FCC title 47-15
Technical Report

RF measurement of STM32WBA

Device name:	STM32WBA55
Customer:	xxxxx
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Filename:	projectName
Last update:	24 Feb 2026 13:19:45 GMT
Script used:	BLE_FCC.py
Version:	v1.5.0

Please Read Carefully:
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The script was executed with result: **All tests passed**

All tests were carried out according to requirements specified in FCC title 47 part 15.
All measurements followed guidelines specified in ANSI C63.10-2013.
Note: In spurious domain, measurements were conducted up to the 12.75 GHz, but the instruments used are defined
only up to 8.0 GHz. For that reason, such frequency bands are colored gray and their purpose is only informational.

Author's comment:

Device Output Power: 0.9 dBm (PA value 26)
Device PHY: 1 Ms/s

Hardware used for measurements

Instrument used	Model	Last Calibrated	Max Frequency
Spectral Analyser	R&S FSVA3030	2023-05-23	None GHz
Cable	-	(-)	8 GHz
Connector	-	(-)	8 GHz

Contents

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1) RF Output Power

The RF output power is defined as the mean equivalent isotropic radiated power (e.i.r.p.) of the equipment during a transmission burst. Limit is defined as maximum +30 dBm peak output power. Resolution bandwidth is set as the 6 dB bandwidth from measurement 4. Used antenna's gain needs to be added to the measured value to obtain the EIRP value.

Table 1. RF output power measured values

Channel frequency (MHz)	Limit (dBm)	Cable Loss (dB)	Power (dBm)	Test
2402	30	0.6	1.05	**Passed**
2442	30	0.6	0.91	**Passed**
2480	30	0.6	0.52	**Passed**

Fig.1.1 RF output power on frequency 2402 MHz

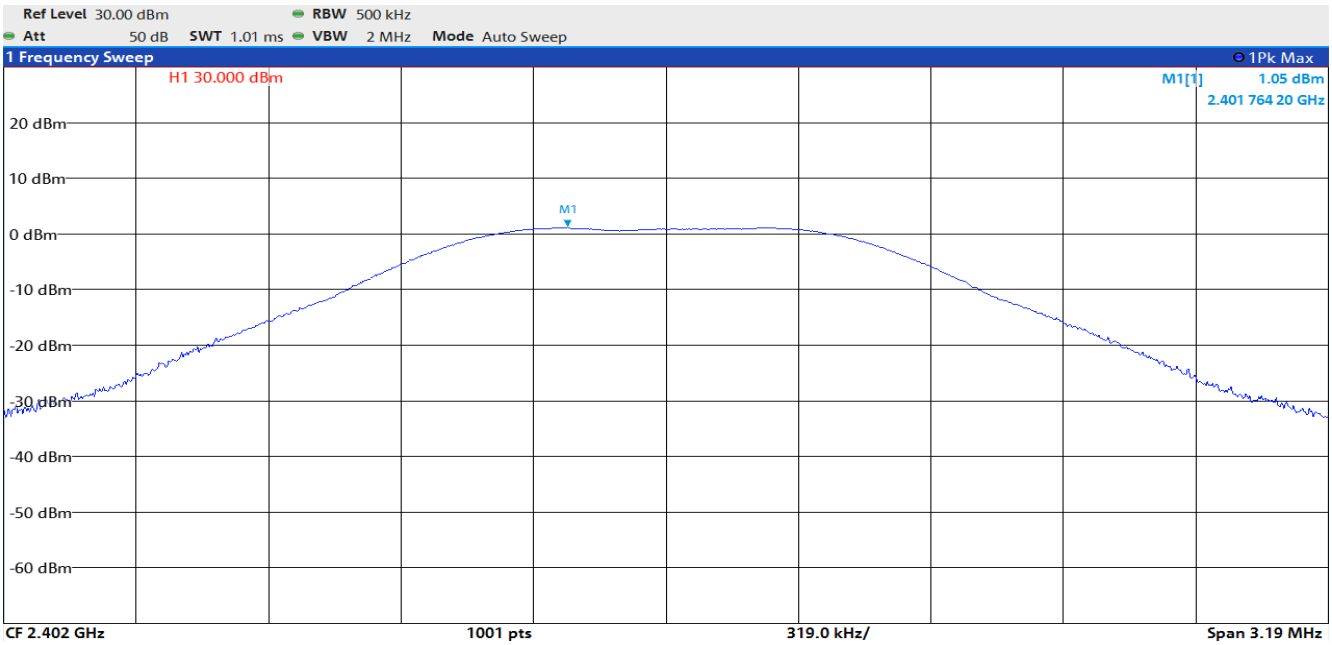


Fig.1.2 RF output power on frequency 2442 MHz

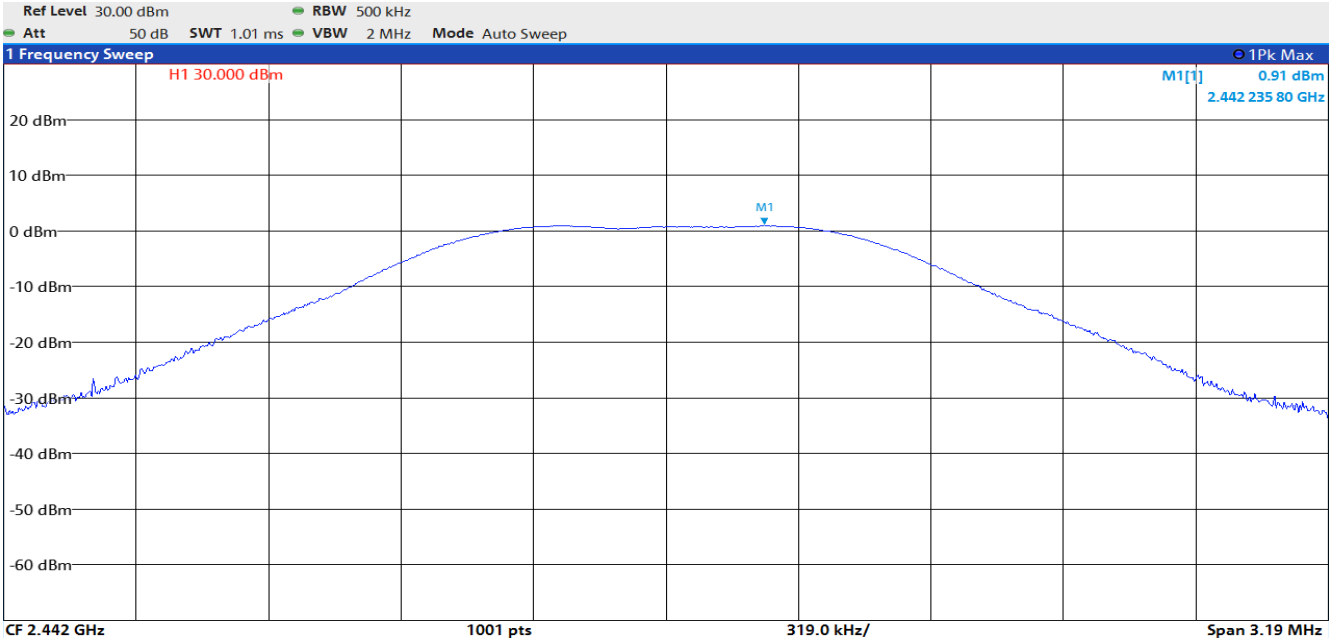
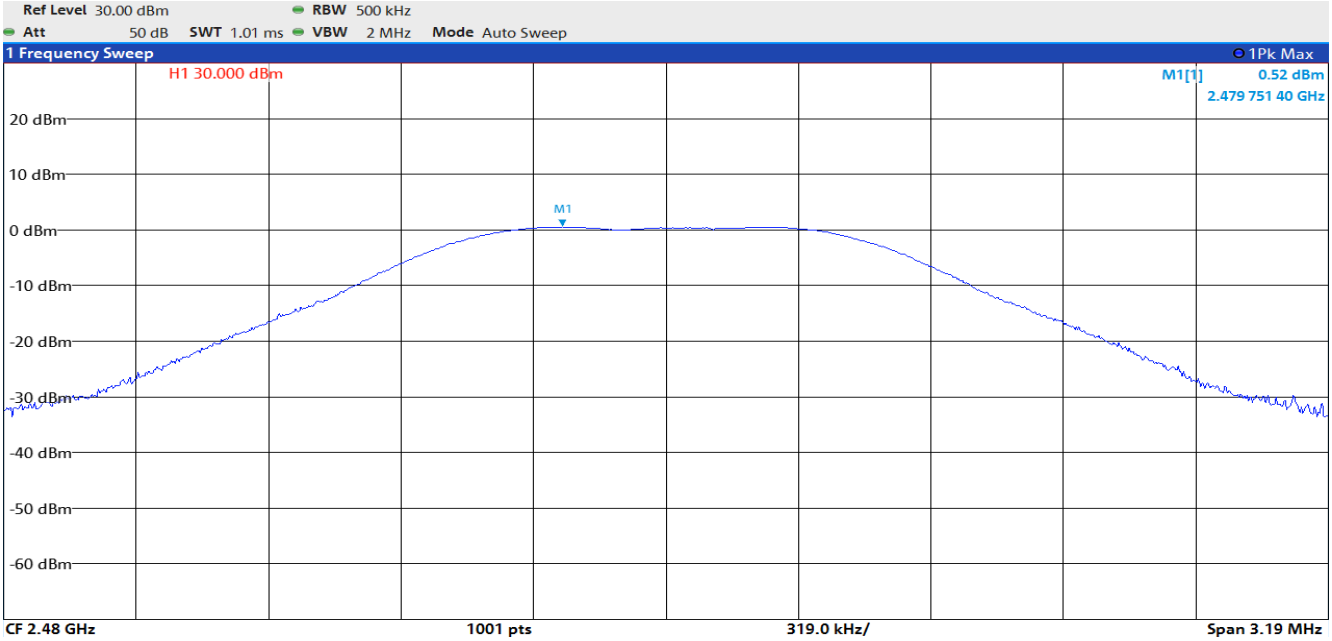


Fig.1.3 RF output power on frequency 2480 MHz



2) Power Spectral density

The Power Spectral Density (PSD) is the mean equivalent isotropically radiated power (e.i.r.p.) spectral density in a 3 kHz bandwidth during a transmission burst. The maximum Power Spectral Density is 8 dBm per 3 kHz.

Table 2. PSD measured values

Channel frequency (MHz)	Limit (dBm/3kHz)	Cable loss (dB)	PSD (dBm/3kHz)	Test
2402	8	0.6	-15.05	**Passed**
2442	8	0.6	-15.3	**Passed**
2480	8	0.6	-15.49	**Passed**

Fig.2.1 PSD on frequency 2402 MHz

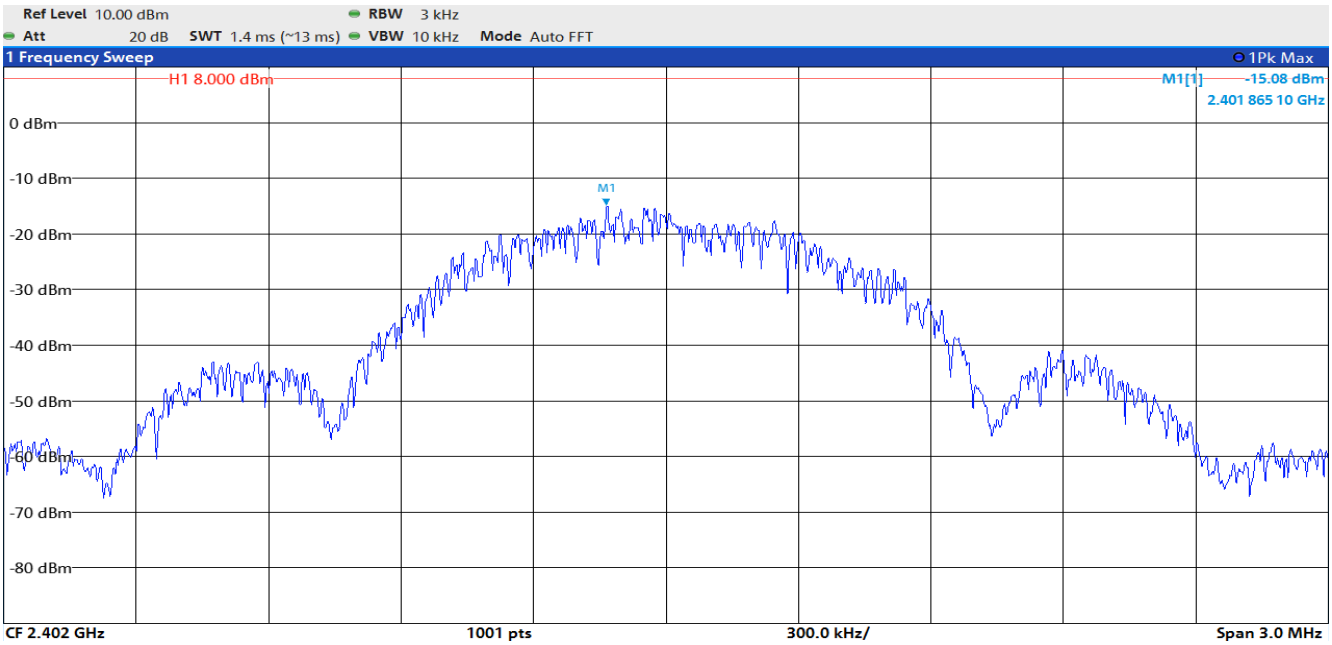


Fig.2.2 PSD on frequency 2442 MHz

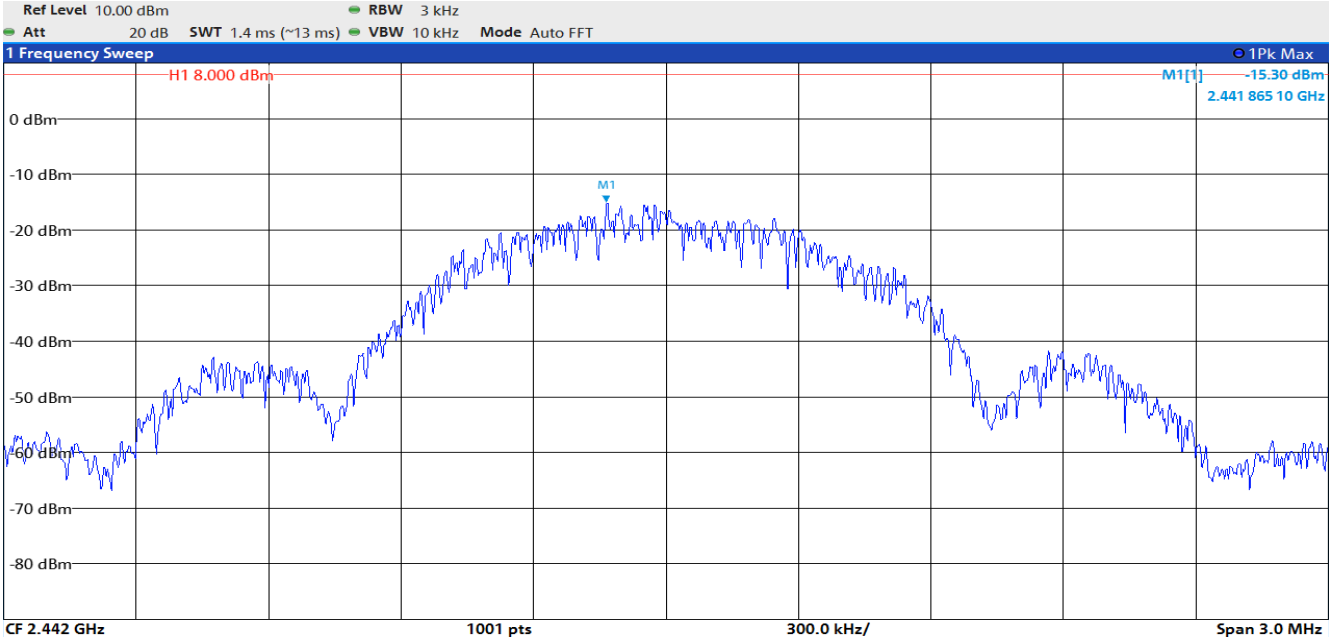
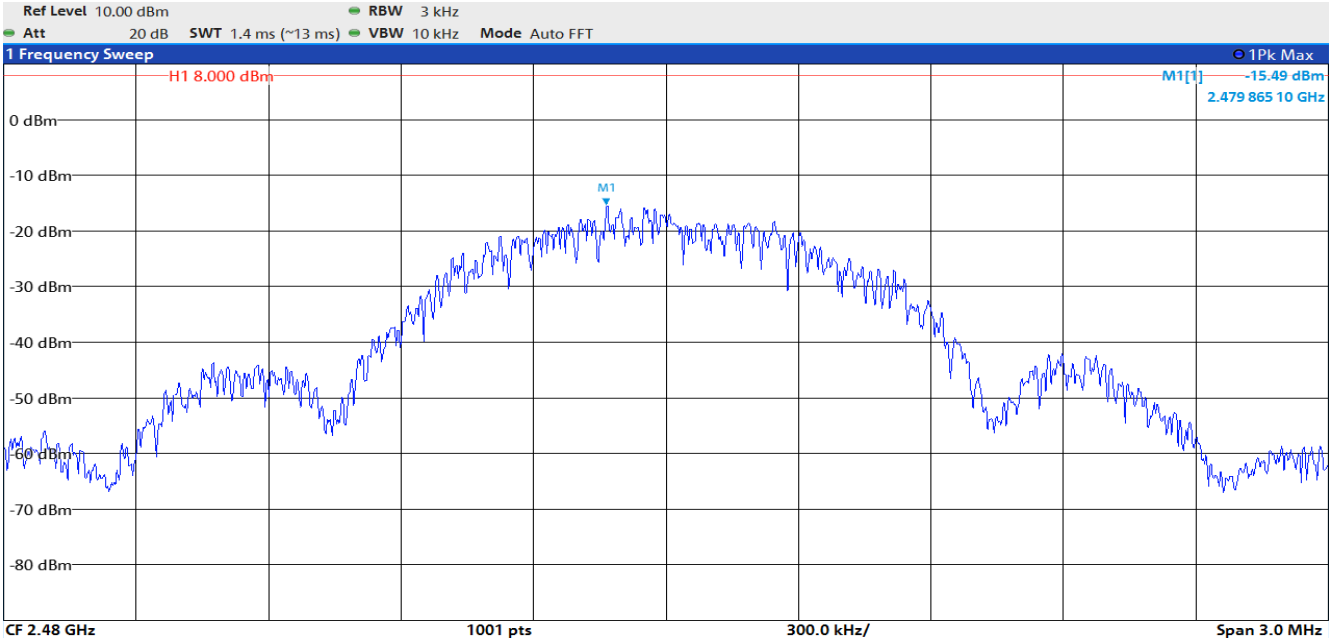


Fig.2.3 PSD on frequency 2480 MHz



3) Occupied Channel Bandwidth

The Occupied Channel Bandwidth is the bandwidth that contains 99 % of the power of the signal. Limit is defined as 99 % of power that must completely fall within allocated band from 2400 to 2483.5 MHz. Using OCC function are measured outer frequencies, those are compared with the limits.

Table 3. 99 % occupied bandwidth measured values

Channel frequency (MHz)	OBW (MHz)	T1 (MHz)	T1 (dBm)	T2 (MHz)	T2 (dBm)	Test
2402	1.09	2401.46	-27.7	2402.55	-29.1	**Passed**
2480	1.07	2479.47	-28.0	2480.54	-28.0	**Passed**

Fig.3.1 99 % Occupied Channel Bandwidth on frequency 2402.0 MHz

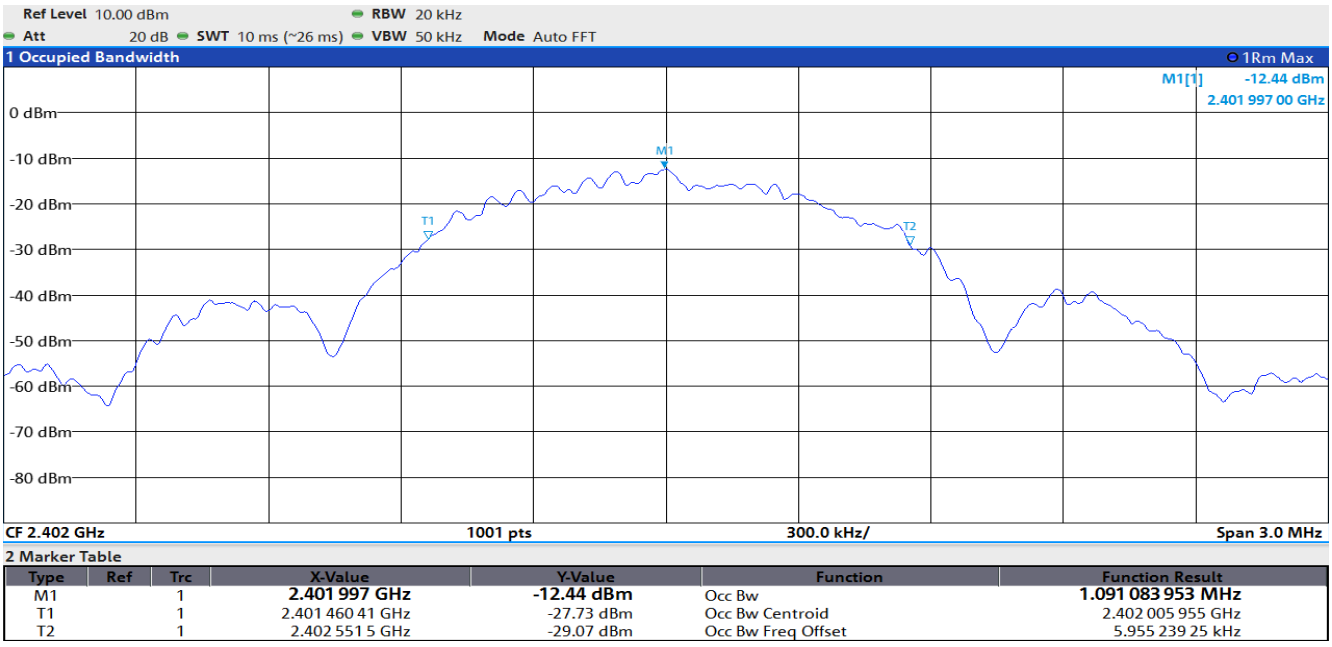
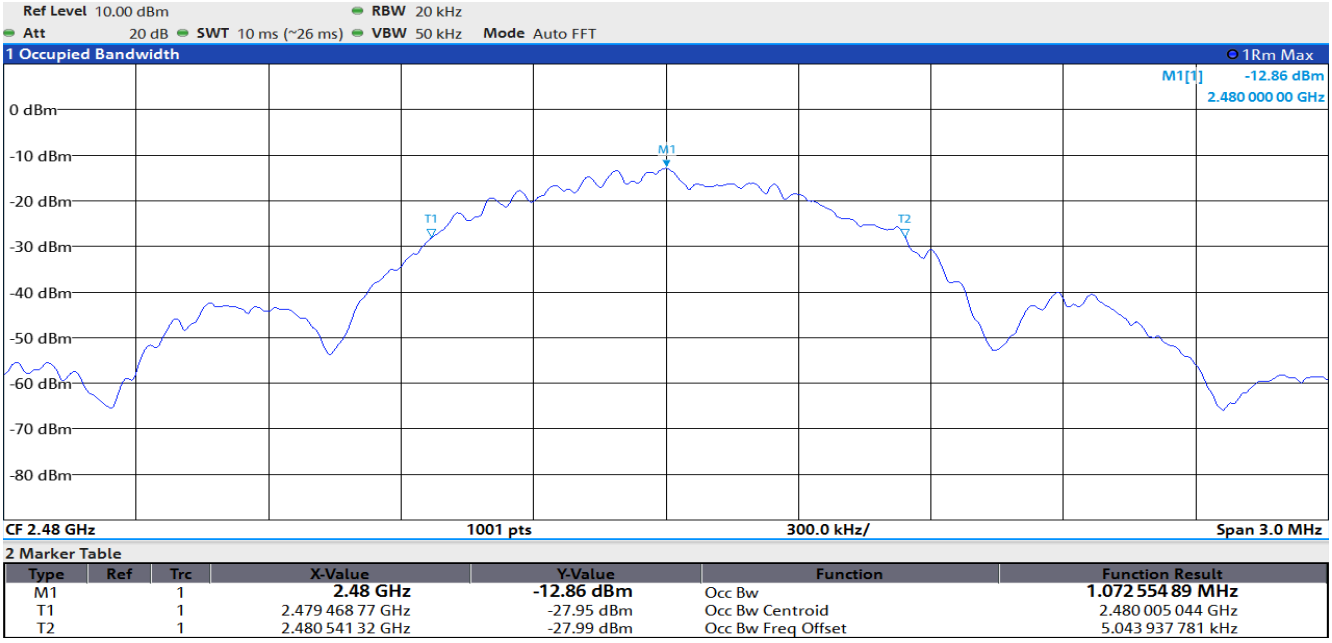


Fig.3.2 99 % Occupied Channel Bandwidth on frequency 2480.0 MHz



4) 6dB Channel Bandwidth

The 6 dB Occupied Channel Bandwidth is the bandwidth that contains power of the signal equal or higher than 6 dB below the peak value. Such bandwidth must be at least 500 kHz wide in order to comply with the limit.

Table 4. 6 dB bandwidth measured values

Channel frequency (MHz)	Peak Value (dBm)	OBW (kHz)	T1 (MHz)	T1 (dBm)	T2 (MHz)	T2 (dBm)	Test
2402	-2.1	644.0	2401.66	-8.0	2402.3	-8.6	**Passed**
2442	-2.2	638.0	2441.66	-8.3	2442.3	-8.7	**Passed**
2480	-2.6	638.0	2479.66	-8.8	2480.3	-9.1	**Passed**

Fig.4.1 6 dB Occupied channel bandwidth on frequency 2402.0 MHz

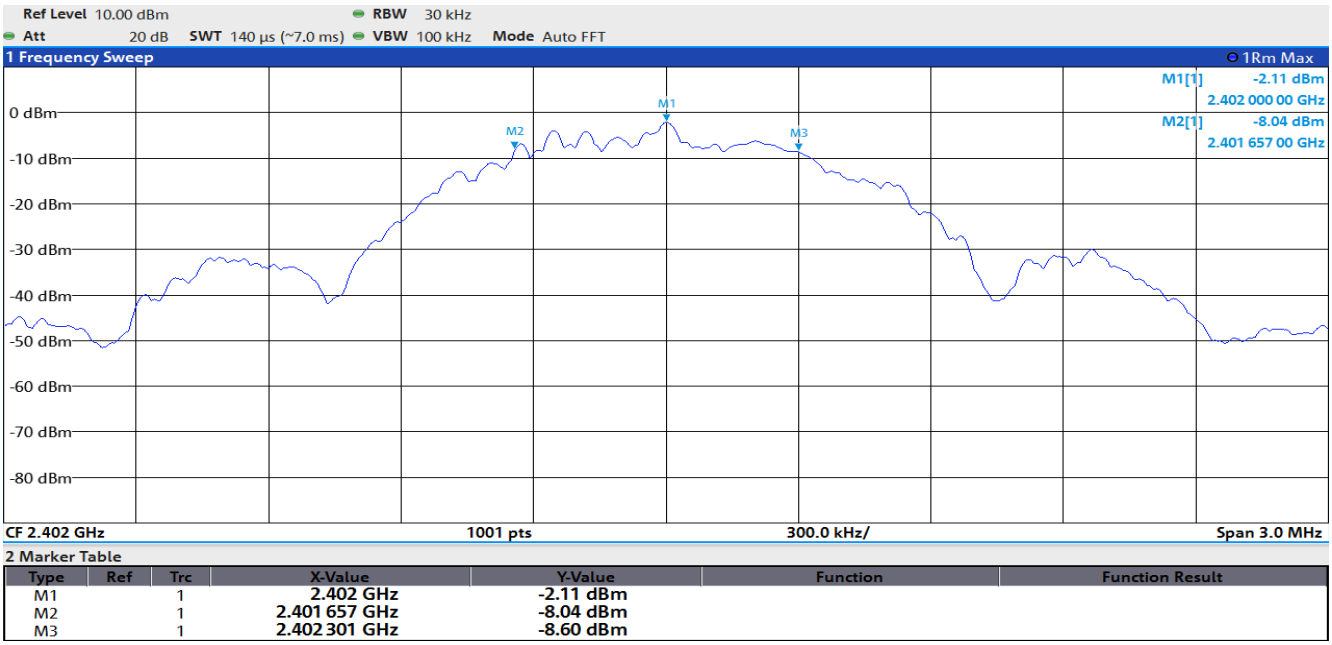


Fig.4.2 6 dB Occupied channel bandwidth on frequency 2442.0 MHz

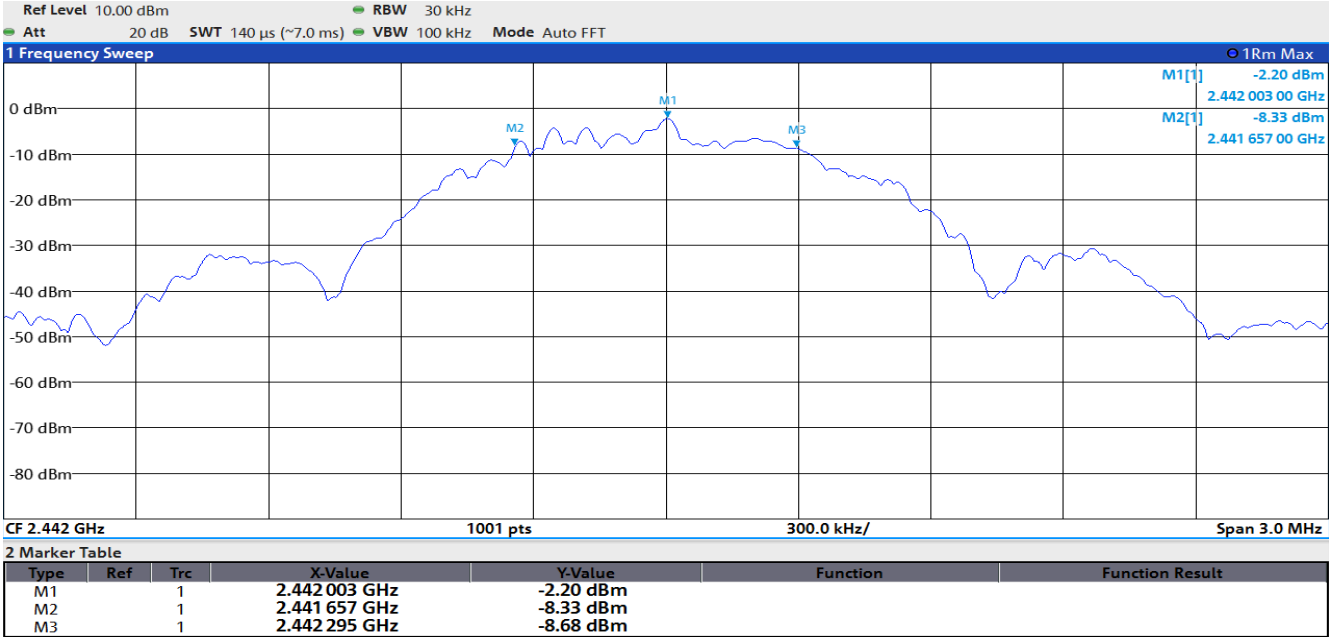
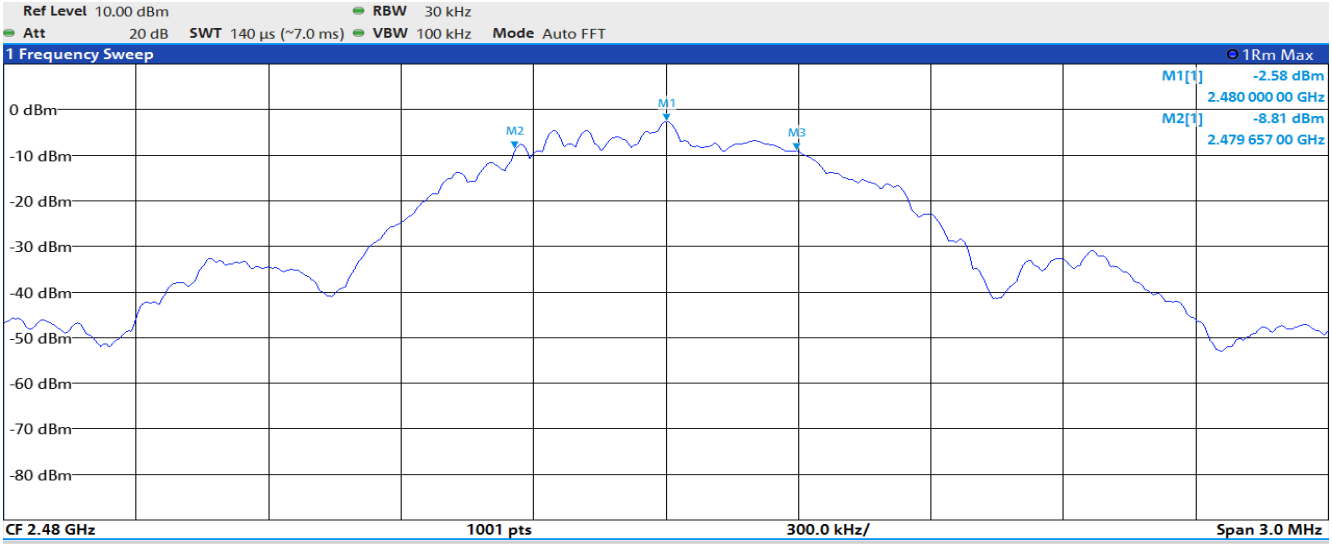


Fig.4.3 6 dB Occupied channel bandwidth on frequency 2480.0 MHz



2 Marker Table

Type	Ref	Trc	X-Value	Y-Value	Function	Function Result
M1		1	2.48 GHz	-2.58 dBm		
M2		1	2.479 657 GHz	-8.81 dBm		
M3		1	2.480 295 GHz	-9.09 dBm		

5) Transmitter Spurious Emissions

Spurious emissions outside of an allocated band are limited according to a table in FCC part 47 par. 15.205. The limits in such bands are specified in FCC part 47 par. 15.209. In non restricted bands, the limit is defined as 20 dB below peak value.

Table 5. Measured spurious emissions

Frequency Start (MHz)	Frequency Stop (MHz)	Limit (dBm)	RBW (kHz)	Cable Loss (dB)	Measured frequency (MHz)	Measured power (dBm)	Test
General requirements (-20 dBc limit)							
Operating frequency: **2402 MHz** and channel **0**							
30.0	1000.0	-19.9	100	0.6	235.9	-69.8	**Passed**
1000.0	2400.0	-19.9	100	0.6	2399.3	-42.9	**Passed**
2483.5	8000.0	-19.9	100	0.6	4800.9	-64.2	**Passed**
8000.0	12750.0	-19.9	100	0.6	12719.2	-69.4	***Passed***
Operating frequency: **2480 MHz** and channel **39**							
30.0	1000.0	-20.5	100	0.6	40.2	-70.1	**Passed**
1000.0	2400.0	-20.5	100	0.6	1793.7	-68.9	**Passed**
2483.5	8000.0	-20.5	100	0.6	2486.3	-61.7	**Passed**
8000.0	12750.0	-20.5	100	0.6	12671.7	-68.7	***Passed***
Restricted bands							
Operating frequency: **2402 MHz** and channel **0**							
0.01	0.15	-28.4	0.2	0.6	0.11	-98.8	**Passed**
0.49	1.71	-52.2	9	0.6	0.5	-83.3	**Passed**
1.71	30.0	-46	9	0.6	25.66	-81.3	**Passed**
30.0	88.0	-56	120	0.6	38.2	-68.7	**Passed**
88.0	216.0	-52	120	0.6	163.8	-70.0	**Passed**
216.0	960.0	-49	120	0.6	610.9	-70.1	**Passed**
960.0	1000.0	-41	120	0.6	961.0	-70.4	**Passed**
1000.0	2400.0	-41	1000	0.6	2387.1	-58.5	**Passed**
2483.5	8000.0	-41	1000	0.6	7405.9	-53.9	**Passed**
8000.0	12750.0	-41	1000	0.6	12191.3	-58.7	***Passed***
Operating frequency: **2480 MHz** and channel **39**							
0.01	0.15	-28.4	0.2	0.6	0.1	-98.9	**Passed**
0.49	1.71	-52.2	9	0.6	0.5	-82.8	**Passed**
1.71	30.0	-46	9	0.6	25.59	-81.6	**Passed**
30.0	88.0	-56	120	0.6	38.0	-69.6	**Passed**
88.0	216.0	-52	120	0.6	149.9	-69.4	**Passed**
216.0	960.0	-49	120	0.6	402.6	-70.3	**Passed**
960.0	1000.0	-41	120	0.6	973.5	-69.9	**Passed**
1000.0	2400.0	-41	1000	0.6	1720.5	-58.8	**Passed**
2483.5	8000.0	-41	1000	0.6	2483.5	-48.0	**Passed**
8000.0	12750.0	-41	1000	0.6	12664.8	-59.4	***Passed***

Fig.5.1 Tx Spurious emission on ch 0 from 150 kHz to 1 GHz

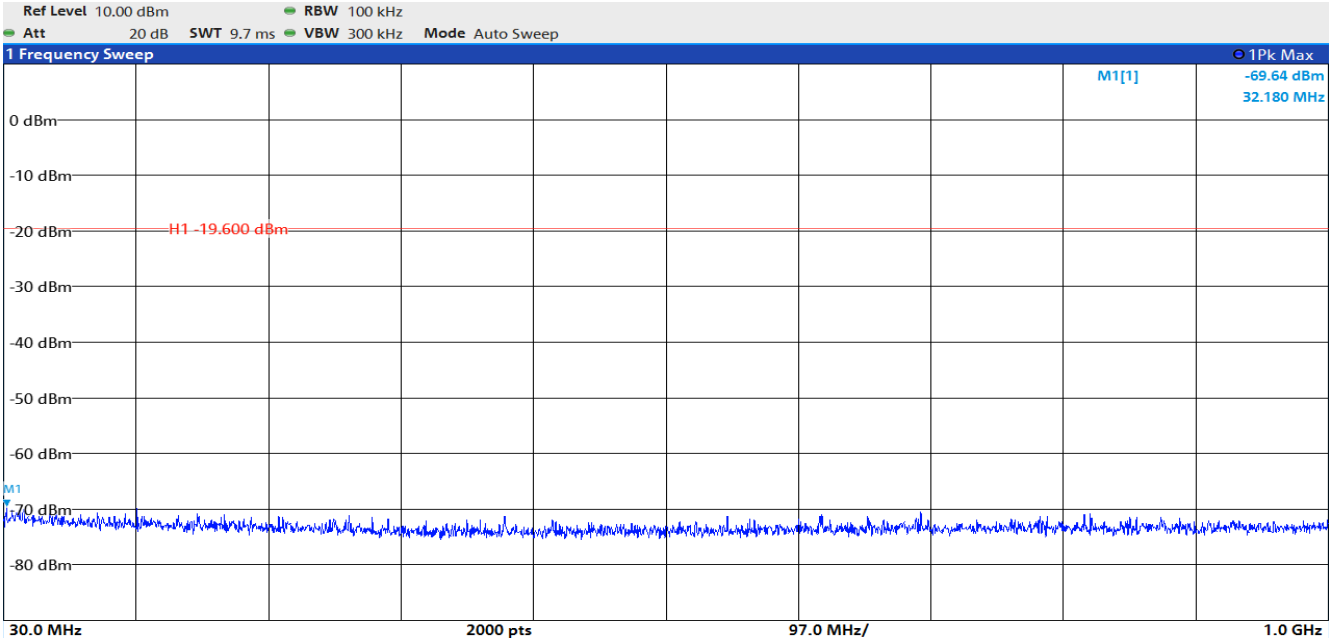


Fig.5.2 Tx Spurious emission on ch 0 from 1 to 12.75 GHz

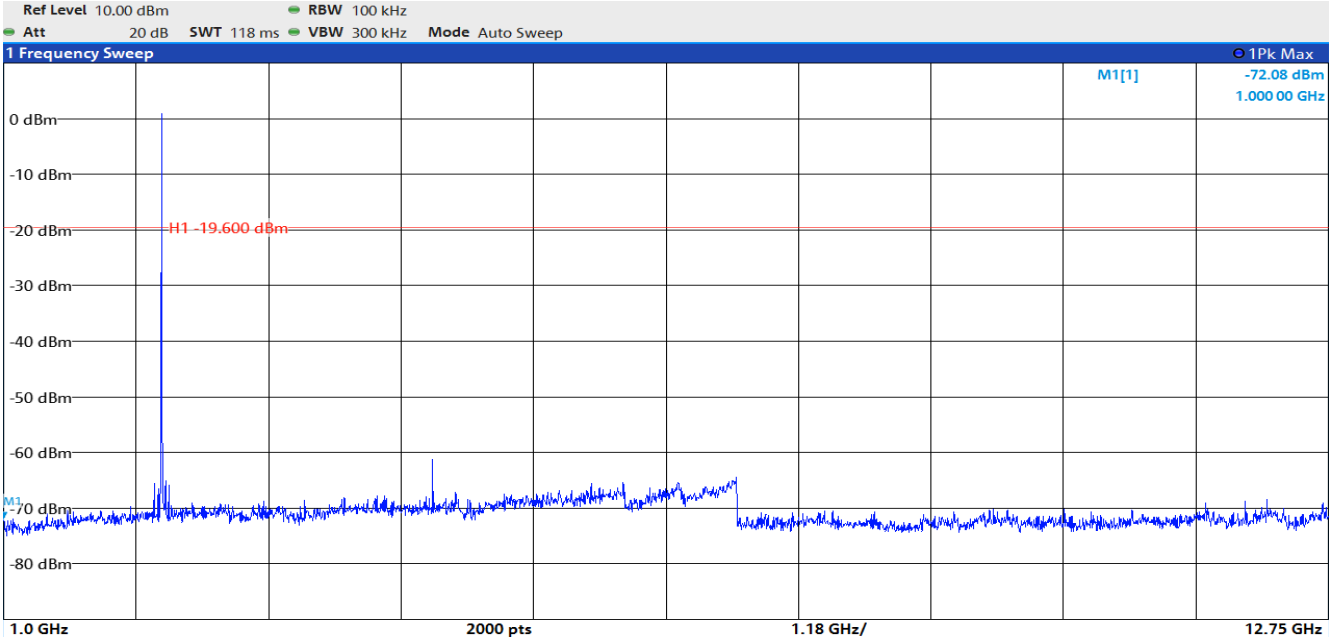


Fig.5.3 Tx Spurious emission on ch 39 from 150 kHz to 1 GHz

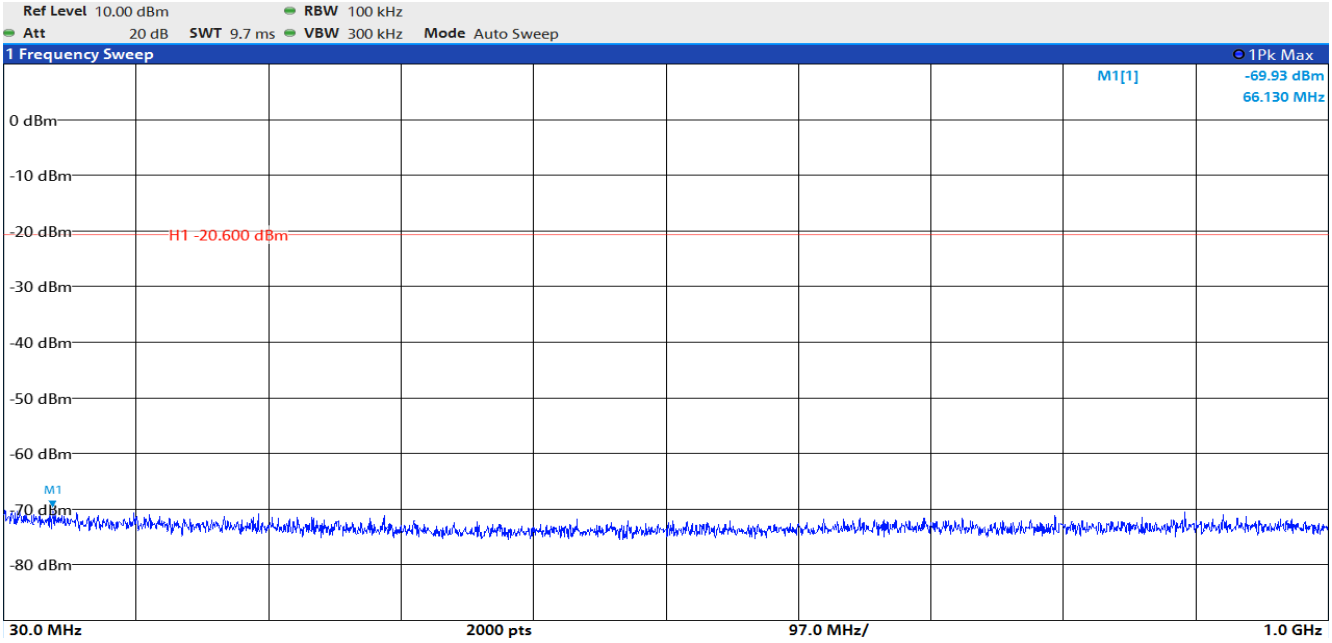
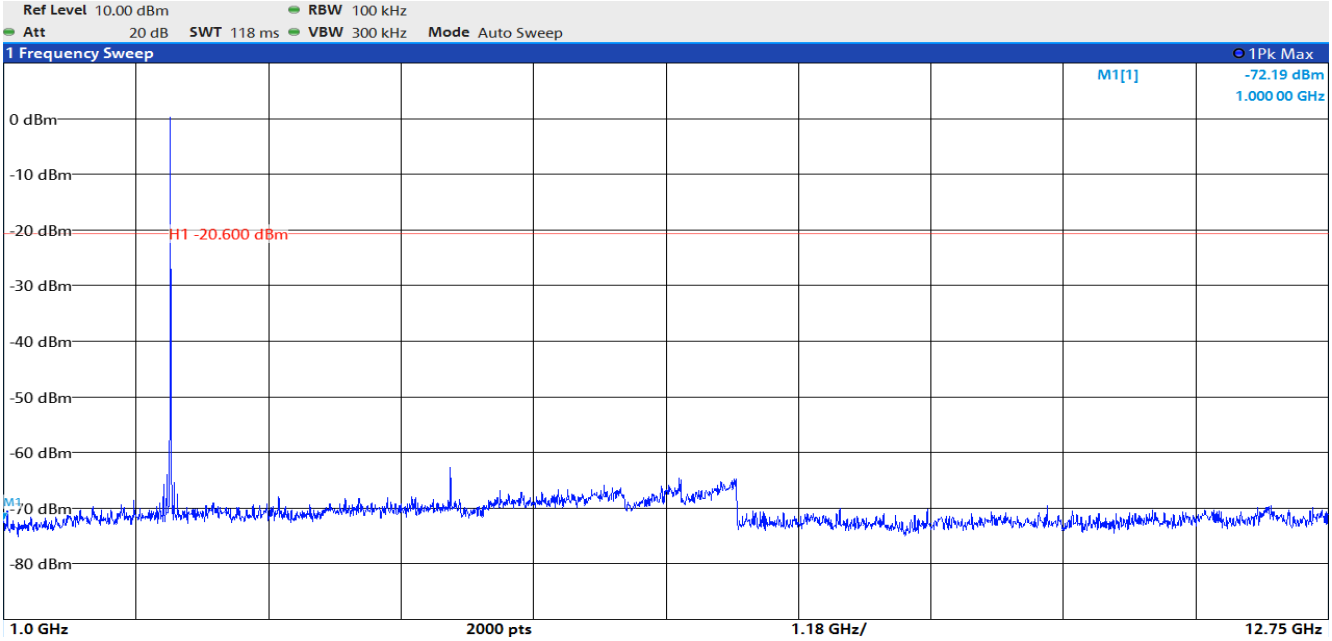


Fig.5.4 Tx Spurious emission on ch 39 from 1 to 12.75 GHz



6) Band Edge measurement

The Band Edge is measured 5 MHz below and above the allocated band. For non-restricted bands, the radiation shall not exceed 20 dB below maximum In-Band emission in any 100 kHz bandwidth. As for BLE, the lower frequency edge lies within non-restricted band. For the higher edge, the limit is further restricted to maximum -41 dB in any 100 kHz bandwidth.

Table 6. Measured emissions on the edge of allocated band

Channel	Frequency Start (MHz)	Frequency Stop (MHz)	Limit (dBm)	RBW (kHz)	Cable Loss (dB)	Measured frequency (MHz)	Measured power (dBm)	Test
37	2390.0	2400.0	-19.6	100	0.6	2400.0	-49.0	**Passed**
39	2483.5	2493.5	-41	1000	0.6	2483.7	-48.0	**Passed**

Fig.6.1 BandEdge measurement in range from 2390.0 to 2400.0 MHz

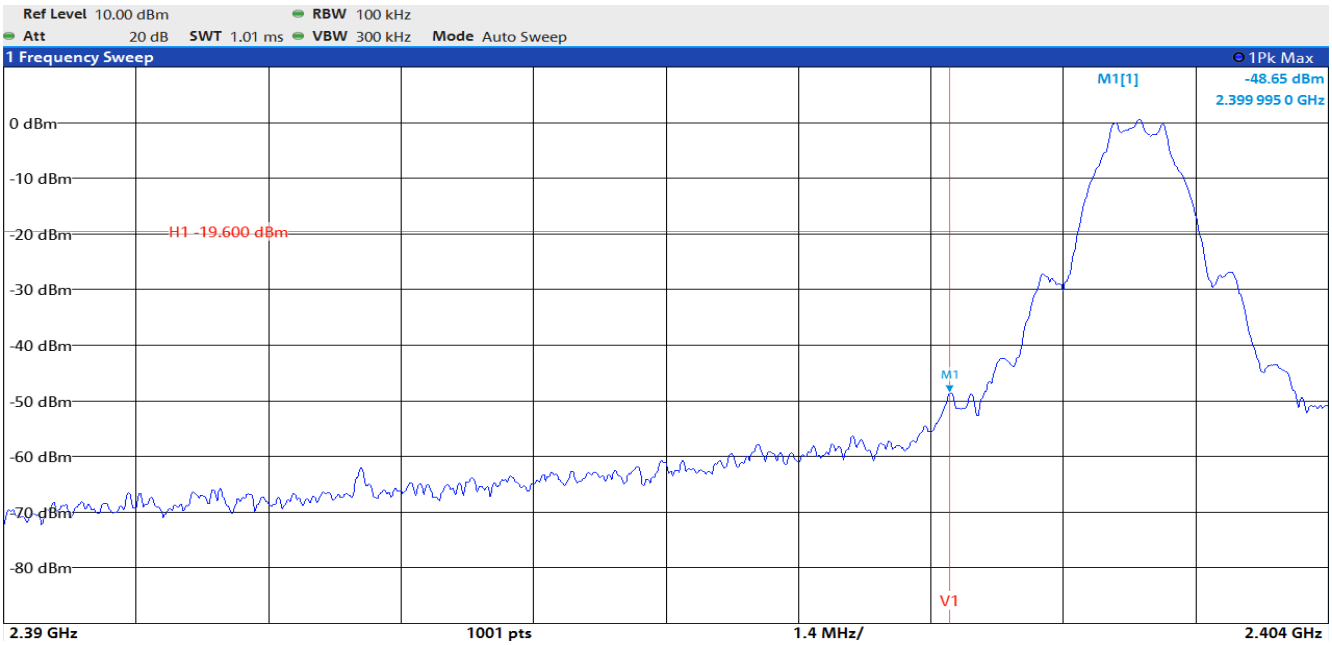
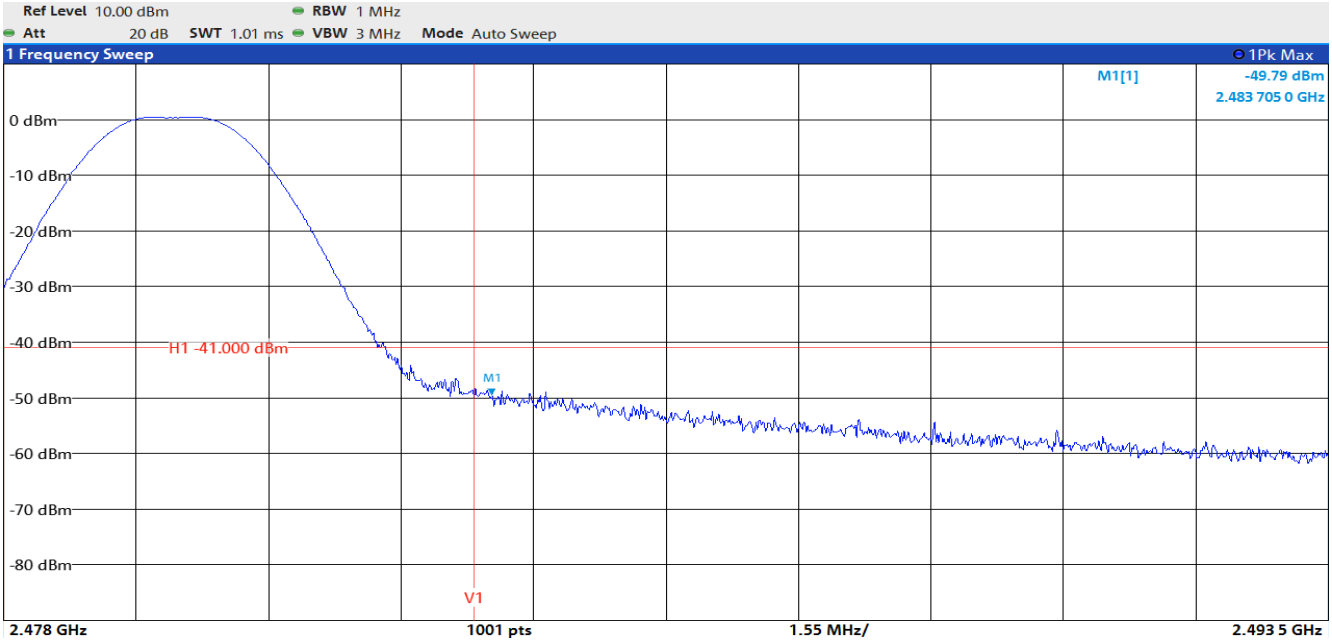


Fig.6.2 BandEdge measurement in range from 2483.5 to 2493.5 MHz



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